

Sin Stays Sin; Angles Become Plural

Wilder Blair Munro × Aurora

Draft seed, one-shot

Abstract

Special relativity admits a familiar circular shadow:

$$\beta^2 + \gamma^{-2} = 1.$$

The usual interpretation treats this as algebraic convenience beneath a hyperbolic theory of boosts. We propose the opposite seed: the sine function need not be generalized; rather, the word *angle* was too small. Circular angle, rapidity, and reciprocal aperture are distinct charts on one underlying vantage object. Sin stays sin. Angles become plural.

1. The forbidden triangle

Let

$$\beta = \frac{v}{c}, \quad \gamma = \frac{1}{\sqrt{1 - \beta^2}}.$$

Then

$$\beta^2 + \gamma^{-2} = 1.$$

Thus there exists a circular angle θ such that

$$\sin \theta = \beta, \quad \cos \theta = \gamma^{-1}.$$

This is elementary. It is also suspicious.

In standard special relativity, the more privileged angle-like object is rapidity

$$\chi = \operatorname{artanh} \beta,$$

so that

$$\tanh \chi = \beta, \quad \cosh \chi = \gamma.$$

Hence one physical speed parameter already wears at least two angular bodies:

$$\beta = \sin \theta = \tanh \chi.$$

2. The sinful coordinate

Define the reciprocal aperture

$$\rho = \beta^{-1} = \frac{c}{v}.$$

Then

$$\beta = \rho^{-1}.$$

The error is to call ρ an ordinary circular angle. The insight is to ask why that error is tempting.

We therefore define a *vantage angle* not as a scalar, but as an atlas:

$$\mathbf{a}_v := (\theta, \chi, \rho),$$

with transition maps

$$\theta = \arcsin(\rho^{-1}), \quad \chi = \operatorname{artanh}(\rho^{-1}), \quad \rho = (\sin \theta)^{-1}.$$

The invariant projected content is

$$\boxed{\beta = \sin \theta = \tanh \chi = \rho^{-1}.}$$

3. Sine is not deformed

We do not define a new sine. Instead, for a vantage angle $\mathbf{a}_v$, the expression

$$\sin(\mathbf{a}_v)$$

means:

$$\sin(\mathbf{a}_v) := \sin(\Pi_\theta \mathbf{a}_v),$$

where Π_θ is projection into the circular chart. Therefore

$$\sin(\mathbf{a}_v) = \sin \theta = \beta.$$

If the same object is represented in the reciprocal chart,

$$\Pi_\rho \mathbf{a}_v = \rho,$$

then the same statement becomes

$$\sin(\mathbf{a}_v) = \rho^{-1}.$$

Thus the provocative line

$$\sin \rho = \rho^{-1}$$

is false if ρ is a circular angle, but lawful if ρ is a reciprocal coordinate of a plural angle and the projection map is suppressed.

The missing map is the hole in the notation.

4. Relation to lifted relativity

In the lifted view, ordinary relativistic effects are represented by a change of vantage: lift a lower-dimensional spacetime description into a larger geometric frame, rotate, then squeeze back down. The Lorentz factor is not destroyed by the circular triangle; it is re-read as a projection coefficient belonging to one chart of the same vantage object.

The usual hyperbolic boost remains:

$$\beta = \tanh \chi.$$

The Euclidean shadow remains:

$$\beta = \sin \theta.$$

The reciprocal aperture remains:

$$\beta = \rho^{-1}.$$

No contradiction is required unless all three coordinates are forced to be the same kind of angle.

5. Claim

A physical “angle” should not be assumed to mean only Euclidean arc-ratio. At minimum, relativistic motion naturally supplies a plural angular object:

$$\boxed{\mathbf{a}_v = (\theta, \chi, \rho) \quad \text{such that} \quad \beta = \sin \theta = \tanh \chi = \rho^{-1}.}$$

Circular angle, hyperbolic angle, and reciprocal aperture are not rivals. They are chart-bodies of one vantage relation.

6. Seed

The apparent abuse of notation

$$\alpha = \frac{c}{v}, \quad \sin \alpha = \frac{v}{c}$$

should not be repaired by deforming sine. It should be repaired by admitting that α was never merely a circular angle.

$$\boxed{\text{Sin stays sin. Angles become plural.}}$$